

The Hand That Measures

The Cupped-Hand Body-Schema, the Palm-and-Handful Measurement Family, the Scoop-and-Ladle Tool Family, and the Etymology of Greek δῶρον 2

Companion paper to The Hand That Gives (Paper 1 of the der- project)

Working paper v2 — for scholarly review

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ABSTRACT

Frisk's *Griechisches Etymologisches Wörterbuch* and Beekes's *Etymological Dictionary of Greek* both treat Greek δῶρον as two entries: δῶρον 1 “gift,” with its established derivation from PIE *deh₃-ro-, and δῶρον 2 “palm-of-the-hand, palm-as-measure,” marked by Frisk as origin unknown and by Beekes (2010: 363) as <[?]> (origin uncertain). The Hesychian gloss δῶρον · παλαιστή, the metrological compounds in *Iliad* Δ 109 and Hesiod *Works and Days* 426, the anatomical δῶρον of Pollux 2.144, and Vitruvius's *De architectura* 2.3.3 all attest the second word. What none of these sources supply is its etymology.

This paper supplies that etymology. δῶρον 2 belongs to a productive PIE body-part-and-tool family built on *de-r- “hand, open palm, cupped hand” — the body-part heteroclite arm-noun developed in the companion paper *The Hand That Gives*. The derivation is Greek-internal: Szemerényi's Law applied to the o-grade nominative *do-r-s yields the lengthened form *dōr (exactly as *eu-patōr-s yields εὐπάτωρ and *deh₃-tōr-s yields Homeric δώτωρ); thematicization then produces *dōr-o-n, which surfaces as δῶρον 2.

The paper's central architectural advance over earlier treatments is the recognition that *de-r- generates a coherent cross-IE family across three distinct measurement-and-form registers, each anchored on a different aspect of the cupped hand:

- The **palm-as-linear-measure** register (palm-breadth, ~3 inches): δῶρον 2, Hesychian δάρις/δάρειρ, Old Irish *derna*, Continental Celtic DERNV[, Albanian *dorë* (developed in a standalone companion paper)
- The **handful-as-volume-measure** register on zero-grade *d̥r-k-: δράξ “handful,” δράσσομαι “to grasp,” δράγμα “sheaf,” δραχμή “drachma” (the monetary handful)
- The **scoop-and-ladle tool** register: Sanskrit *darvī* “ladle, cobra's hood” (lexicalizing the cupped configuration itself), Luwian *taruwal-* “mortar,” Hittite *tarwāli-* “pestle/grinder,” Armenian *targal/dgal* “ladle, spoon” — with parallel Sanskrit-internal hand-as-spoon compounds *dāruhasta-* and *pāṇikā* as typological witnesses

A parallel cluster on the s-stem sister heteroclite *dōws- generates Sanskrit *dos* = 18 inches (*Mātaṅgalīlā* 1.14) and *dor-jyā* “arm-string” = sine (*Sūryasiddhānta* 3.10), placing these forms inside the broader cross-IE cubit family (Latin *cubitus*, Greek πῆχυς, Hebrew *ammah* “mother of the arm,” Old English *ell*).

The organizing principle underlying all three registers is a body-schema: the cupped hand as the body's primal scoop. The arm-and-cupped-hand configuration generates, on one hand, tool-extensions — the ladle is a frozen arm-and-cupped-hand — and, on the other, measurement registers at different scales: palm-breadth for linear

measure, cupped-hand volume for volumetric measure, forearm-length for cubit-class measure. This schema explains why a single root family produces such a tight semantic cluster across the IE branches, and why mainstream etymology has repeatedly reached for “substrate” to explain forms whose phonology is irregular but whose semantic field is unmistakable.

The framework does not contest δῶπον 1 or its standard derivation. Vitruvius’s etymology — “the Greeks call the palm δῶπον because giving is called δῶπον” — is folk etymology of the ordinary kind, a semantic bridge constructed across two homophones. The Celtic fist-cluster (*durno-, OIr *dorn*, Welsh *dwrn*, Breton *dorn*) is reassigned to a separate root family (*d^her-/ *d^hergh- “hold firm, bind tight, knot”) and treated in the companion paper *A Firm Hand*.

Probability conditional on Paper 1’s framework: 50–60%. Project-level marginal probability: 2–5%.

1. Introduction: The Two δῶπα

Greek δῶπον presents a puzzle that the standard etymological dictionaries have not resolved. Frisk’s *Griechisches Etymologisches Wörterbuch* (1960–72) lists two entries: δῶπον “Gabe, Geschenk” (gift) and δῶπον “Handfläche, Palme” (palm of the hand), the second marked “Origin: XX [etym. unknown],” with a side note flagging the comparison to Albanian *dorë* via La Piana. Beekes (2010: 363) preserves the same two-entry structure — δῶπον 1 “gift” with its standard derivation from PIE *deh₃-ro-, and δῶπον 2 “palm-breadth, palm-of-the-hand” marked <?> — and his ETYM section surveys rejected comparanda without arriving at a positive reconstruction. The two-entry treatment is not a framework innovation; it is the position the standard dictionaries already hold, and it leaves the second word’s history open.

The lexicographic record for δῶπον 2 is ancient and varied. The Hesychian lexicon glosses it as παλαιστή — “palm-breadth,” the width of four fingers held together. The word appears in compound numerals: *Iliad* Δ 109 has ἑκαδεκάδωρος “sixteen palm-widths long” (describing Pandaros’s bow horns), Hesiod *Works and Days* 426 has δεκάδωρος “ten palm-widths” (of an axle), Pollux *Onomasticon* 2.144 records δῶπον as an anatomical term for the front of the palm, and inscriptions from Miletus preserve it as a standard architectural unit. The form is well-attested across literary, lexicographic, anatomical, and epigraphic registers from the eighth century BCE onward.

The only ancient attempt to explain the homophony comes from Vitruvius (*De architectura* 2.3.3): the Greeks call the palm δῶπον because giving — *datio* — is called δῶπον. This is folk etymology in its standard form, the kind of story any writer would supply when confronted with two identical-sounding words. The semantic link is not implausible — the HAND → GIVE grammaticalization pathway documented by Heine and Kuteva (2002) is real — but as derivational evidence it explains nothing. English *ear* “organ of hearing” and *ear* “spike of grain” are synchronically identical yet etymologically unrelated; folk etymology links them anyway. The same applies to the two δῶπα.

This paper takes the two-entry structure at face value and fills in the etymology Frisk left blank. δῶπον 1 descends from the give-verb *deh₃- via the nominalization *deh₃-ro-, an analysis the framework adopts without modification. δῶπον 2 descends from the body-part heteroclit **de-r-* of *The Hand That Gives*. The same root generates Hesychian δάρις “span,” Old Irish *derna* “palm,” Continental Celtic DERNV[(Stifter 2022), and Albanian *dorë* (developed in a standalone companion paper).

What version 2 adds beyond the earlier presentation is the recognition that **de-r-* is productive across three distinct registers, all anchored on the same body-schema — the cupped hand and the arm-and-cupped-hand configuration:

1. The **palm-as-linear-measure** register (§4), generating δῶπον 2, δάρις, *derna*, DERNV[, and *dorë*

2. The **handful-as-volume-measure** register on zero-grade *d_ṛ-k- (§5), generating δράξ, δράσσομαι, δράγμα, δραχμή
3. The **scoop-and-ladle tool** register (§6), generating Sanskrit *darvī*, *dāruhasta*-, Luwian *taruwal*-, Hittite *tarwāli*-, and Armenian *targal/dgal*

These are not three unrelated developments; they are three specializations of the same primal body-configuration, lexicalized differently in different daughter languages. The body-schema is articulated in §3.

Three late-twentieth-century departures dismantled the Pokorny IEW 203 grouping that had united these forms. Schrijver (1995: 58) called the δωρ-/δαρ- relation “obscure.” Beekes (2010) excluded the Celtic forms on the grounds that Celtic *u “excludes any relationship with δῶρον.” Matasović (2009) classified Celtic *durno- as a probable substrate borrowing. The first of these objections is addressed in §4.5. The second and third target the fist-cluster (*durno-, *dorn/dwrn*, Breton *dorn*, Latvian *dūre*), which the framework reassigns to PIE *d^her-/ *d^hergh- in *A Firm Hand*, removing the Celtic *u problem from the present paper entirely.

The paper proceeds as follows. §2 sets out the formal foundation: Szemerényi’s Law, thematicization, and the Greek-internal model. §3 articulates the cupped-hand body-schema. §4 develops the palm-as-linear-measure cluster. §5 develops the handful-as-volume-measure cluster on zero-grade *d_ṛ-k-. §6 develops the scoop-and-ladle tool cluster. §7 develops the arm-as-cubit parallel on the s-stem sister *dóws-. §8 frames the body-part heteroclite class as a productive PIE pattern. §9 presents the full etymology of δῶρον 2 with derivation table. §10 addresses δῶρον 1 and the Vitruvian folk etymology. §§11–12 cover the coexistence reading and the late-twentieth-century departures. §§13–15 address scope, falsifiability, and probability.

2. The Formal Foundation

The derivation of δῶρον 2 and its open-palm cluster siblings rests on a formal apparatus that is entirely Greek-internal and draws on sound laws and morphological processes uncontroversial in the standard handbooks. Three steps operate in sequence: Szemerényi’s Law on the consonant-stem r-stem nominative singular; thematicization of the resulting athematic stem; and, for the volume-measure cluster, zero-grade morphology with velar extension.

2.1 Szemerényi’s Law and the εὐπάτωρ/δῶτωρ Pattern

Greek preserves a productive class of r-stem nouns and adjectives in -ωρ whose long nominative vowel descends from Szemerényi’s Law operating on pre-PIE *-Vrs# sequences. The exemplars are uncontroversial. The adjective εὐπάτωρ “of a noble father” (accusative εὐπάτορα) descends from *eu-patōr-s, the nominative ending *-s triggering compensatory lengthening and loss of the sibilant: pre-PIE *-Vrs# → PIE *-VR̥#. The Homeric agent-noun δῶτωρ “giver” (*Od.* 8.335) descends from *deh₃-tōr-s by the same mechanism. The pattern runs throughout the Greek r-stem and t-stem agent-noun systems: ρήτωρ “speaker,” σωτήρ “savior,” πατήρ “father,” μήτηρ “mother,” θυγάτηρ “daughter.”

Szemerényi’s Law is among the best-established sound rules in historical IE phonology. Lundquist and Yates (2018: 2083) formulate it as “PIE word-final *-VRs# → *-VR̥#” — a regular phonological rule on athematic consonant-stem nominatives whose stem-final element is a sonorant followed by *-s. Standard examples: *ph₂tér “father” from *ph₂tér-s, *k(u)wōn “dog” from *k(u)wén-s (Szemerényi 1996: 113–119). Applied to the r-stem *de-r- with o-grade nominative *do-r-s, the law produces *dōr by exactly the mechanism that produces εὐπάτωρ from *eu-patōr-s; the e-grade *de-r-s yields *dēr in parallel.

2.2 Thematicization of Athematic R-Stems

The lengthened nominative *dōr produced by Szemerényi’s Law is a bare athematic root-noun. Greek inherits a productive process by which such athematic stems are reanalyzed as bases for thematic derivation, yielding neuter o-stem nouns in *-o-m/*-o-n alongside or in place of the inherited athematic form — a process documented in every reference grammar of historical IE morphology (Beekes 2011: 211–214; Fortson 2010: 132–134; Meier-Brügger 2003: 219–225).

Greek-internal parallels are plentiful. δόρυ “spear, beam, tree-trunk” (genitive δόρατος) is a heteroclitic neuter that survives in attested Greek both as an athematic u-stem and as a thematicized form in compounds and derivatives. γόνυ “knee” (genitive γόνατος) shows the same pattern. For δῶρον specifically, the Mycenaean form *do-ra* (KN Sd 4401, in the chariot tablets) attests δῶρα as a thematicized neuter plural already in the thirteenth century BCE, confirming that the surface morphology was in place by the Mycenaean period.

2.3 Zero-Grade Morphology and Velar Extension

A third mechanism is required for the volume-measure cluster developed in §5. PIE *d_ṛ- (zero-grade of *de-r-) extended by the velar suffix *-k- yields the Greek δράξ / δράσσομαι / δράγμα / δραχμή family. The α-vocalism is the regular Greek outcome of syllabic *ṛ (compare δράκων, πατρός, ἀνδρός). Zero-grade *d_ṛ-k- is morphologically parallel to many other zero-grade nominal formations in Greek and across IE; it involves nothing exotic. The morphology is developed in §5.

2.4 The Morphological Class

The framework’s apparatus rests on the standard handbook treatment of Szemerényi’s Law and the productive thematicization of r-stems, plus standard zero-grade nominal morphology. Beekes’s *Etymological Dictionary of Greek* (2010: 363) marks δῶρον 2 <?> and surveys rejected comparanda without supplying a positive reconstruction; Beekes is a witness for the unresolved status of the etymology, not a source of support for the framework’s *de-r- derivation.

3. The Body-Schema Anchor: The Cupped Hand as Primal Scoop

This section articulates the body-schema that organizes the rest of the paper. The schema is not itself a derivational claim; it is a conceptual anchor that explains why a single PIE root generates such a coherent semantic cluster across the daughter languages.

3.1 The Cupped Hand as Primal Scoop

Before measuring vessels, before ladles, before coined money, there was the cupped hand. Humans used it to drink from streams, to scoop grain, to carry small quantities of material. The cupped hand is the body’s primal scoop: a real, named, three-dimensional configuration with a concave shape (open palm) and a natural function (scooping, containing, measuring) that is among the most ancient and universal of bodily activities.

The framework’s central claim is that the PIE root *de-r- has its semantic anchor in the cupped hand and its function as primal scoop, container, and measure. From this configuration, three semantic registers emerge in the daughter languages:

- The cupped hand’s *form* (palm and span as linear extents) → linear-measure register (palm-as-measure)
- The cupped hand’s *contents* (handful as volume) → volume-measure register (handful-as-measure)
- The cupped hand’s *function* (scooping, containing) → tool register (ladle, scoop, mortar)

These are not three separate semantic developments but three aspects of a single body-configuration foregrounded differently in different lexical specializations. The cupped hand is simultaneously a shape (palm-breadth), a container (handful-volume), and a tool-prototype (the body’s own ladle-bowl).

Running orthogonally to the three registers is a second organizing axis: the body-schema branches into two productive lexicalization channels in the daughter languages.

- The **body-part channel** lexicalizes the hand-as-body-part. The word's referent is anatomical: a palm, a span, a hand. Outputs include Greek δῶρον 2 “palm,” Greek δάρις “span,” Old Irish *derma* “palm,” Continental Celtic DERNV[, and Albanian *dorë* “hand.”
- The **configuration channel** lexicalizes the cupped/scoop *shape* abstracted from the hand. The word's referent is the shape itself, applicable to any object that shares it — including objects that are not hands at all. Sanskrit *darvī* is the clearest case: it applies to ladles (cupped wooden tools), to the splayed hood of a cobra (a cupped shape on a vertical column with no anatomical hand involved), and to certain plants with characteristic concave leaves. The Anatolian mortar-and-pestle vocabulary and Armenian *targal/dgal* “ladle” belong here as well.

Both channels are productive outputs of a single body-schema. They are not competing etymologies but sister lexicalizations, foregrounding either the body-part itself or the shape it produces. The same morphology (e-grade r-stem with i-stem extension) can output a body-part word in one daughter language (Greek δάρις = “span”) and a configuration word in another (Sanskrit *darvī* = “cupped scoop-shape”); the morphological apparatus is identical, but the daughter-language lexicalization selects which channel surfaces.

3.2 The Arm-and-Cupped-Hand Configuration

The cupped hand does not exist in isolation; it sits at the end of an arm. This arm-and-cupped-hand configuration is the body's gestalt for “extended limb with concave terminus” — and it has a direct artifactual instantiation: the ladle.

A ladle is structurally the arm-and-cupped-hand configuration frozen in wood or metal. The shaft corresponds to the forearm; the bowl corresponds to the cupped hand; the wrist joint corresponds to the joint between shaft and bowl. This is not a metaphor that requires interpretive generosity; it is the most direct possible body-to-artifact correspondence. The Sanskrit compound *dāruhasta-* “wood-hand = wooden spoon” (§6.2) explicitly preserves this identity within IE: Sanskrit speakers used “hand” as the head-noun of the compound for “spoon,” with “wood” as the material modifier. The ladle is a frozen arm-and-cupped-hand; the body and the artifact are conceptually one object named with the same lexical resources.

3.3 Two Heteroclite Members on One Body-Schema

The body-part heteroclite class developed in *The Hand That Gives* has two members: the r-stem *de-r- and the s-stem *dōws- (the arm-noun). On the body-schema reading these are not distinct body-parts but focal variants of the same body-configuration, foregrounding different aspects of the arm-and-cupped-hand gestalt:

- *de-r- foregrounds the **distal terminus** — the cupped hand, the palm, the handful, the scoop
- *dōws- foregrounds the **proximal shaft** — the arm, the forearm, the cubit-as-shaft-measure

Both heteroclites name the same underlying configuration; they differ in which aspect the lexicon brings to the front. The metrology they generate reflects this: the r-stem yields palm/span-class measures (~3–9 inches), the s-stem yields cubit-class measures (~18 inches). Each heteroclite generates measurement vocabulary at the scale appropriate to the aspect of the arm-and-cupped-hand it foregrounds.

3.4 Cross-Linguistic Typological Support

The body-schema reading is supported by a well-documented cross-linguistic typology. Heine and Kuteva's *World Lexicon of Grammaticalization* (2002) identifies HAND → CONTAINER as one of the productive cross-domain grammaticalization channels. Hilpert (2007) documents the metonymic chains by which body-part terms acquire new readings through habitual use. Pattillo (2014) identifies the cupped-hand-as-grain-measure

across multiple language families (Sanskrit *añjali*, Latin *manipulus*, Persian *kapithi*, Hebrew *kaph*). The framework’s body-schema for *de-r- sits within the typology these scholars document.

4. The Palm-as-Linear-Measure Cluster on *de-r-

The cognate forms lexicalized as the open palm or palm measurement — Hesychian δάρις/δάρειρ, Old Irish *derna*, Continental Celtic DERNV[, and δῶρον 2 itself — all belong to the palm-as-linear-measure register of the *de-r- family. The natural output of this register is a measurement unit of palm-breadth scale (~3 inches).

4.1 Hesychian Greek δάρις / δάρειρ (σπιθαμή)

The Hesychian lexicon glosses δάρις · σπιθαμή — “span,” the distance from the tip of the extended thumb to the tip of the extended little finger. The standard σπιθαμή equals 12 fingers or half a cubit (Herodotus 2.106). Chantraine (1968: 305) gives the dialectal distribution: δάριν is Arcadian, δάρειρ its Laconian counterpart with the regular Laconian rhotacism. The open-palm cluster thus surfaces in three distinct Greek dialectal traditions: Arcadian δάριν, Laconian δάρειρ, and the broader Attic-Ionic and Doric δῶρον 2.

Morphologically, δάρις is an i-stem derivative on the bare e-grade r-stem *de-r-, the arm-stem extended with the i-stem suffix and lexicalized as the measurement unit of the spread hand. Frisk (GEW I 430–431) treats δάρις/δάρειρ on the same footing. Chantraine groups δάριν, δάρειρ, and δῶρον 2 with Albanian *dorë* under a common etymon, citing Pokorny IEW 203. The Hesychian gloss survives in the marginal apparatus because the literary tradition adopted χεῖρ as the standard “hand” word, but measurement terminology is conservative, and the body-part anchor persisted in it.

4.2 Old Irish *derna* (Gaelic *deàrna*) “Palm”

Old Irish *derna* (Gaelic *deàrna*) “palm of the hand” presents alternant nominative *dernu* and genitive *dernand*. Pedersen (1913, vol. II: 110) places *dernu/dernand* in the same morphological paradigm as *gualu/gualand* “shoulder,” *Ánu/Ánann* (the goddess), and *áru/árann* “kidney” — a Celtic-internal classification among alternating-stem nouns, made six years before Marstrand published. The form is built on the bare e-grade r-stem *de-r- with a feminine singulative *-nā- extension, lexicalized specifically as the open palm. The Celtic palm-meaning is one of the daughter-language ARM → HAND specializations documented in *The Hand That Gives* §4.3.

4.3 Continental Celtic DERNV[at the Bronze of Novallas

The Celtiberian *hospitium* bronze published by Stifter (2022) preserves the form DERNV[in a Continental Celtic epigraphic context, anchoring the open-palm cluster in Continental Celtic epigraphy. Stifter notes explicitly that the relationship between Old Irish *dernu* “palm” and *dorn* “fist” cannot be a simple case of ablaut — consistent with the framework’s reassignment of *durno- to a separate root family in *A Firm Hand*. The relationship between *derna* and *durno- is not ablauting at all; it reflects two different PIE roots that happen to converge phonologically in some Celtic outputs.

The same bronze preserves the Celtiberian directional preposition DV, the Celtiberian reflex of the Pre-PIE directional particle *de/do developed in *The Hand That Gives* §3.2. Two outputs of the framework’s bare *de- — the body-part noun on the r-stem and the directional particle on the bare stem — survive in a single inscription.

4.4 δῶρον · παλαιστή: The Body-Part Sense (δῶρον 2)

The Hesychian gloss δῶρον · παλαιστή places “palm-breadth” alongside the gift-noun under the same lemma. The body-part and metrological senses are attested from the eighth century BCE forward: ἑκκαδεκάδωρος “sixteen palm-widths long” in *Iliad* Δ 109; δεκάδωρος “ten palm-widths” in Hesiod *Works and Days* 426; the anatomical record of Pollux 2.144; and the architectural inscriptions from Miletus. LSJ groups all of these

under δῶρον II (palm-related), distinct from δῶρον I (gift). The full derivation from *do-r-s via Szemerényi and thematicization is developed in §9.

4.5 The δῶρ-/δάρ- Relation: Sibling Derivations on *de-r-

Schrijver (1995: 58) called the δῶρ-/δάρ- relation “obscure”: the long ὀ of δῶρον 2 and the short α of δάρυς cannot be connected by any straightforward sound rule. Schrijver’s verdict is correct — under the assumption that δῶρ- and δάρ- stand in direct phonological relationship. The framework rejects that assumption.

δῶρον 2 and δάρυς are not phonological variants of a single form; they are sibling derivations on a shared base *de-r-. δῶρον 2 is the thematicized output of the Szemerényi-derived o-grade nominative *dōr (from *do-r-s); δάρυς is the i-stem extension on the bare e-grade *der-. Different morphological operations on different ablaut grades of the same root. The forms cannot be derived from each other precisely because they are not variants — they are independent derivations from the same stem at different grades.

Three morphological operations on the same base yield three distinct outputs in the palm-as-linear-measure register: δῶρον 2 (Greek, o-grade, Szemerényi-lengthened, thematicized), δάρυς (Greek, e-grade, i-stem), *derna* (Celtic, e-grade, feminine singulative). Albanian *dorë* is a fourth: e-grade, Szemerényi-lengthened, *-eh₂- feminine extension (Chavis and Claude 2026c). Schrijver asked the right question of the wrong data. The question is not “what sound rule links δῶρ- and δάρ-?” but “what derivational scaffold relates δῶρον 2, δάρυς, *derna*, and *dorë*?” The framework supplies that answer at the morphological level, where the relation is transparent.

The zero-grade of the same base (*d_r-) generates the volume-measure cluster of §5 — δράξ, δράσσομαι, δράγμα, δραχμή. The three measurement registers (linear palm, volumetric handful, cubit-class arm) are sibling-derivational outputs on one root family, distinguished by ablaut grade and suffix selection.

5. The Handful-as-Volume-Measure Cluster on Zero-Grade *d_r-k-

This section develops the cluster lexicalized as the cupped-hand-volume measure: Greek δράξ “handful,” δράσσομαι “to grasp,” δράγμα “sheaf, that which is grasped,” and δραχμή “drachma.” The cluster sits on the zero-grade of *de-r- extended by a velar suffix, and supplies the volumetric-measure register parallel to the linear-measure register of §4. Where δῶρον 2 names the palm as a flat shape (palm-breadth, ~3 inches), δράξ names the cupped hand as a container (the handful, the volume of grain that fits in a closed palm).

5.1 Greek δράξ “Handful”

δράξ (genitive δρακός) “handful, fistful, as much as one can grasp in the hand” is the standard Greek word for the handful as a unit. It descends from PIE *d_r-k- (zero-grade of *de-r- with velar extension), the α-vocalism reflecting the regular Greek outcome of syllabic *r̥ (compare δράκων from *d_rk-ōn, πατρός from *pH₂t_r-os, ἀνδρός from *h₂n_r-os). The morphology is parallel to other zero-grade nominal formations in Greek and involves nothing unusual.

Beekes (2010) marks δράξ as origin-uncertain and considers a Pre-Greek substrate source. The framework’s reading — zero-grade *d_r-k- of *de-r- — is phonologically at least as defensible and has the substantial advantage of explaining the semantic clustering with the rest of the *de-r- family, rather than treating it as coincidence. Under the framework’s analysis, the α-vocalism that motivates Beekes’s substrate reading falls out as regular zero-grade Greek vocalization.

5.2 δράσσομαι “To Grasp”

δράσσομαι “to grasp, seize, take a handful” names the cupped-hand-closing action: the active grasping motion by which an open cupped hand transitions to a closed fist around contained material. The verb is built on the same zero-grade *d_r-k-/*d_r-g- base. Its semantic core is exactly the body-action that produces a δράξ:

the hand closes around what it has scooped, yielding both the act (the grasping motion) and the result (the handful-as-volume).

5.3 δράγμα “Sheaf, That Which Is Grasped”

δράγμα “sheaf, bundle of stalks, that which is grasped in one hand” is the harvest-noun on the same root, attested from Homer (*Iliad* Σ 552, Δ 472, Λ 69). The semantic is concrete and agricultural: the quantity of grain stalks the harvester closes one hand around as a single unit. The cupped-hand-volume measure appears here in its most immediate agricultural realization — the handful as the natural harvest-unit.

5.4 δραχμή “Drachma, Monetary Handful”

δραχμή is the standard Greek monetary unit, and its name preserves the body-volume anchor explicitly. The drachma was originally a handful of ὀβελοί — iron spits used as proto-currency. Six ὀβελοί made one δραχμή, a quantity defined by what the closed hand could grasp. When coinage replaced spit-currency, the unit-name survived; the etymology, however, still carries the body-volume origin.

The chain runs: body-part name (cupped hand) → measurement unit (handful) → economic unit (drachma). This same pathway appears across languages — English *pound* was originally a weight; Latin *talentum* a weight-measure; Hebrew *shekel* a weight — but δραχμή is unusual in preserving the body anchor so transparently in its very form. The drachma is etymologically “a handful,” and its name documents that the cupped-hand-volume measure was productive in Greek economic life, not merely a poetic image.

5.5 The Cupped-Hand-as-Volume Metrology

The δράξ family belongs to a near-universal pattern: across language families and historical periods, the cupped hand has served as the primal grain-measure. A cross-linguistic sample:

- **Persian *kapithi*** (Greek καπίθη) “a small Persian grain-measure equal to about two Attic *choenikes*” — from Old Iranian *kap- “to grasp, hold,” cognate with Latin *capiō*, Greek κάπτω. Xenophon (*Anabasis* 1.5.6) records that one *kapithi* of wheat-meal cost four *sigli* in Persian-area markets. The unit-name means etymologically “what the hand grasps.”
- **Sanskrit *añjali*** “the two cupped hands brought together as a measure of grain or water” — productive in Vedic ritual and Classical literature as a small dry-measure unit.
- **Latin *manipulus*** “handful; a small military unit of soldiers” — the military sense derived by analogy: a manipule is as many soldiers as the standard-bearer can hold in one hand, parallel to the handful-of-grain measure.
- **Latin *pugillus*** “a small handful, fistful” — productive in Roman pharmacy and small-quantity trade.
- **Hebrew *kaph*** “palm; the hollow of the hand; cupped-hand-measure” — used for measured quantities in biblical and rabbinic literature.

The cupped-hand-as-volume measure is among the most archaic and universal of human measurement categories. The body provides the instrument; the cupped configuration provides the unit. The δράξ family lexicalizes this category in Greek on the same root family that produces the linear palm-measures of §4 and the scoop-tool family of §6.

5.6 The Three Measurement Registers in Greek

Greek lexicalizes all three measurement registers on the same root family, in different ablaut grades with different morphological extensions:

- **Linear, palm-breadth scale:** δῶρον 2 (o-grade Szemerényi + thematicization), δάρις (e-grade i-stem)
- **Volumetric, handful scale:** δράξ (zero-grade velar extension)
- **Linear, cubit scale:** via the parallel s-stem heteroclit **dōws-* (§7)

The cross-IE distribution of these registers preserves the same three-way pattern with different lexical fillings in each branch. This is the framework’s central architectural claim: *de-r- and its s-stem sister *dóws- generate a coherent body-part-and-measure family across IE, daughter languages preserving different focal variants in different lexical slots.

6. The Ladle and Scoop Tool Cluster on *de-r-

This section develops the artifact-cluster generated by the cupped-hand body-schema: the ladle, scoop, mortar, pestle, and related implements of the Indic-Anatolian-Armenian zone. The framework’s claim is that these artifacts are named not from the wood of which they are typically made, but from the body-configuration they preserve — the cupped hand and the arm-and-cupped-hand gestalt. The wood etymology that mainstream IE etymology has applied to several of these forms (via PIE *dóru “wood”) does not survive close scrutiny.

6.1 Sanskrit *darvī* “Ladle, Cobra’s Hood”

Sanskrit *darvī* (दर्वी) is attested from the *Atharvaveda* forward with two principal senses: “ladle, large spoon” (RV V, X; AV; productive throughout Vedic and Classical Sanskrit ritual vocabulary) and “the splayed hood of a cobra” (AV X.4.13). The compounds *darvī-kara* “hood-bearing snake” and *vi-darvya* “hoodless serpent” confirm that the bare form *darvī* carries the meaning “hood, cupped/spread shape” independently of the ladle sense. Crucially, *darvī* in Sanskrit never refers to a hand. Its semantic core, recoverable from its full attested range, is the cupped or splayed concave configuration itself — the shape, abstracted from any anatomical referent.

This is the framework’s central case for the configuration channel. The PIE *de-r- “cupped hand” body-schema branches in Sanskrit not as a body-part word (Sanskrit has *hastā-* and *pāṇi-* for “hand”), but as a configuration word: *de-r- with feminine i-stem extension on the e-grade lexicalizes the cupped shape itself, transferable to any object presenting that shape. Morphologically the form is parallel to Greek δάρυς (both feminine i-stems on bare e-grade *de-r-), but Greek lexicalizes the body-part sense as a span-measurement, while Sanskrit lexicalizes the configuration sense as a shape-word.

The cobra-hood attestation is diagnostic. A cobra in striking posture presents a vertical column — the raised body — surmounted by a splayed concave hood. This shape is silhouette-identical to a vertical forearm with a cupped hand at the top; it is in no other sense a hand. Sanskrit speakers naming both the ladle and the cobra-hood with the same word *darvī* were responding to a body-schema correspondence that is visually and geometrically real but does not pass through any body-part terminology. The shape is the referent.

The mainstream etymology derives *darvī* from *dāru- “wood” (Wiktionary glosses the form as “wooden one” → “ladle”). This handles the ladle sense but leaves the cobra-hood sense as inexplicable homonymy: a snake’s hood is not wooden, and no semantic path from “the wooden one” reaches “the splayed concave shape on a vertical column.” The configuration reading on *de-r- handles both senses with one mechanism: the word names the shape, and the shape is shared. The cobra-hood is the diagnostic that decides between the two etymologies in favor of the body-schema.

6.2 The Parallel Hand-as-Spoon Lexicalizations in Sanskrit

Independent of the *darvī* question, Sanskrit preserves a separate set of spoon-words built transparently on body-part terms. The compound **dāruhasta-** (दरुहस्त) “wooden spoon, ladle” combines *dāru-* “wood” with *hasta-* “hand” — literally “wood-hand,” with the compound functioning as a noun for “wooden spoon.” Alongside it, *dāruhastaka-* (with diminutive *-ka-*) and *pāṇikā* “kind of spoon” (built on *pāṇi-* “hand” with feminine suffix) document the same compositional pattern from different lexical bases.

These compounds establish that the HAND-as-SPOON conceptual mapping is lexically productive in Sanskrit. Speakers used *hasta-* and *pāṇi-* — the standard body-part terms for “hand” — as head-nouns of compounds for “spoon” and “ladle,” with material modifiers (“wooden”) added compositionally. The body-schema that links cupped hand to scoop-shape is active in Sanskrit not only through the configuration channel (*darvī* as cupped shape) but also through the body-part channel applied compositionally (*hasta-/pāṇi-* + qualifier as spoon-words).

This parallel structure illuminates the relationship between *darvī* and *dāruhasta-*. They are not derivationally related — *dāruhasta-* is built on *hasta-* and *dāru-*, not on *darvī* — and they do not generate semantic conflict. Each lexicalizes the spoon through a different aspect of the same body-schema: *darvī* names it by the cupped shape it shares with the body-configuration; *dāruhasta-* names it by direct reference to the body-part it resembles, with the material specified. Sanskrit thus runs the body-part channel and the configuration channel side by side in its spoon vocabulary, and the existence of both is mutually compatible.

The typological observation supports the body-schema reading of the broader cluster: a language that compositionally constructs spoon-words from hand-words (*dāruhasta-*, *pāṇikā*) and that also lexicalizes the cupped configuration as an abstract shape-word (*darvī*) is a language in which the body-schema linking hand to scoop is conceptually active. This is exactly the conceptual environment the framework reconstructs for PIE.

6.3 Luwian *taruwal-* “Mortar”

Luwian *taruwal-* is the Anatolian mortar-word, attested in cuneiform Luwian ritual texts. The form is structurally close to *de-r- with a tool-extension suffix; the Anatolian reflex of PIE *d as plain *t* is regular (Hittite/Luwian show consistent devoicing of the obstruent series).

The mortar is conceptually a fixed cupped hand: the concave bowl that holds material while the pestle grinds it. In the body-schema, the mortar lexicalizes the cupped-hand-as-container frozen in stone or wood as an artifact. That Near Eastern mortars were typically stone rather than wood is the second piece of evidence against the wood etymology: calling a stone mortar “the wooden one” requires substantial semantic drift. The framework’s reading — the mortar named for the cupped-hand shape of its bowl — requires none.

6.4 Hittite *tarwāli-* “Pestle, Grinder”

Hittite *GIŠtarwāli-* is the pestle-word in cuneiform Hittite ritual and culinary texts, written with the *GIŠ* (“wooden”) determinative. The form is parallel to Luwian *taruwal-* and shows the same Anatolian sound-correspondences to *de-r- with tool-suffix.

The *GIŠ* determinative records what the object is made of, not what the word etymologically means. Hittite pestles were often wooden — for grinding spices and grain — so the determinative reflects practical reality. But the material of an artifact does not determine the etymology of its name. The framework’s reading — that *tarwāli-* names the pestle by the body-action it enacts (the cupped-hand-as-grinder) rather than its material — is entirely consistent with the Hittite ritual vocabulary.

6.5 Armenian *targal/dgal* “Ladle, Spoon”

Old Armenian *targal* / *drgal* / *dgal* (with variants across the manuscript tradition) is the Armenian ladle-and-spoon word, surviving directly into Modern Armenian *dgal* “spoon.” Martirosyan (2010) notes its position in the Anatolian-Armenian contact zone: the word is either an Armenian inheritance from PIE *de-r- with productive tool-suffix, or an Anatolian borrowing on the same root. Either way, the Armenian form preserves the Indic-Anatolian pattern at its westernmost point: Sanskrit *darvī* → Anatolian *taruwal-/tarwāli-* → Armenian *targal/dgal*, all on the same body-schema, all in different daughter-language specializations.

6.6 The Indic-Anatolian-Armenian Zone

The geographic concentration of the configuration-channel tool-cluster is striking. The primary attestations — Sanskrit *darvī*, Luwian *taruwal-*, Hittite *tarwāli-*, and Armenian *targal/dgal* — all sit in the eastern wing of the

IE family, in the zone of extensive cross-branch contact and shared cultural vocabulary. The Sanskrit-internal hand-as-spoon compounds *dāruhasta-* and *pāṇikā* (§6.2) provide additional typological witness from the same zone, documenting that the body-schema linking hand to scoop was conceptually live in Sanskrit alongside the configuration-channel lexicalization.

The framework’s claim is bounded to this zone. It does *not* claim Germanic *trog* “trough” or Old Irish *drochta* “wooden basin” for the same root family. These western-IE wooden-vessel terms are productively derived from PIE **deru-*/**drū-* “tree, wood” in mainstream treatments, and may genuinely belong there. The positive claim is scoped to the eastern zone, where the evidence is strongest: the cobra-hood sense of *darvī*, the parallel hand-as-spoon compositional pattern, the Anatolian mortar and pestle that are not necessarily wooden in material, and the Armenian survival into the modern language.

6.7 The Configuration Channel in the Eastern Zone

The configuration-channel lexicalization of **de-r-* generates four daughter-language outputs in the eastern zone, each naming an artifact by the cupped or concave shape it shares with the primal body-configuration rather than by the body-part it resembles or the material it is made of:

1. **Sanskrit *darvī*** (i-stem on e-grade): the cupped scoop-shape itself, lexicalized across ladles, cobra hoods, and concave plant forms
2. **Luwian *taruwal-*** (with tool-suffix): the cupped-shape mortar (concave receiving-bowl)
3. **Hittite *tarwāli-*** (parallel Anatolian form): the pestle companion to the mortar
4. **Armenian *targal/dgal*** (Armenian, surviving into the modern language): the cupped-shape ladle/spoon

The configuration channel runs alongside the body-part channel (Greek *δόρυ* 2, Albanian *dorë*, Old Irish *derna*, etc.) — both are productive outputs of the same body-schema in different daughter-language lexicalizations. The configuration channel dominates in the eastern zone; the body-part channel dominates in the western and Greek-Albanian-Celtic branches. The eastern zone additionally preserves the body-part channel compositionally (*dāruhasta-*, *pāṇikā* in Sanskrit; §6.2), so the two channels coexist within the same branch where the evidence is densest.

6.8 The Wood-Etymology Problem

Mainstream IE etymology places the eastern ladle/scoop cluster under PIE **dóru* “tree, wood,” glossing Sanskrit *darvī* as “the wooden one” → “ladle” and the Anatolian forms as “wooden vessel” or “wooden tool.” The framework finds this etymology insufficient on two independent grounds.

First, **the cobra-hood problem**: Sanskrit *darvī* “splayed cobra’s hood” (AV X.4.13) cannot descend from “the wooden one.” A snake’s hood is not wooden, and no semantic path leads from “wood” to “splayed concave shape on a vertical column.” The wood etymology accounts for one attested sense and leaves the other inexplicable. The configuration reading on **de-r-* handles both senses with one mechanism: the word names the cupped shape, and the shape is shared between the ladle and the snake.

Second, **the Anatolian stone-mortar problem**: Near Eastern mortars were typically stone rather than wood. Naming a stone mortar “the wooden one” requires substantial semantic drift; the configuration-channel reading requires none — the mortar is named for the shape of its concave receiving-bowl, irrespective of material. The GIŠ (“wooden”) determinative on Hittite *tarwāli-* records what the object was made of, not what the word etymologically meant.

The framework’s PIE **de-r-* “cupped hand” derivation resolves both difficulties cleanly. The Sanskrit-internal hand-as-spoon compounds (*dāruhasta-*, *pāṇikā*; §6.2) provide additional typological support for the broader body-schema: a language that constructs spoon-vocabulary compositionally from hand-words is a language in

which the conceptual link between cupped hand and scoop is lexically live, exactly as the framework reconstructs for PIE.

7. The Arm-as-Cubit Parallel: The s-Stem *dóws- and the Cubit Family

The measurement clusters of §§4–6 all sit on the r-stem *de-r-. The same body-part heteroclite class includes a sister s-stem *dóws- “arm, fore-arm” (Pokorny IEW 226), which generates a parallel measurement cluster at a different scale. The r-stem yields palm/span-class and cupped-hand-volume measures; the s-stem yields cubit-class measures (~18 inches). Metrology is a productive feature of the class as a whole, not an isolated property of either heteroclite.

7.1 Sanskrit *dos* = 18 Inches (*Mātaṅgalīlā* 1.14)

Sanskrit *dos* is attested as a unit of 18 inches in the *Mātaṅgalīlā* 1.14, Nilakaṇṭha Sastri’s elephant-treatise tradition. The 18-inch value corresponds to the standard cubit — the length from elbow to fingertip — and aligns with the cross-cultural pattern of forearm-as-measurement-unit. The form *dos* is the bare s-stem arm-noun (sandhi-form *doḥ*, compound stem *dor-*) of Sanskrit *doṣ-/doṣán-* “arm, fore-arm” developed in *The Hand That Gives* §4.4. The cubit-class measure is the natural metrological output of an arm-noun: the arm noun generates the arm-length measure.

7.2 *dor-jyā* “Arm-String” = Sine (*Sūryasiddhānta* 3.10)

Vedic *dor-jyā* “arm-string” is the Sanskrit mathematical term for the sine in the *Sūryasiddhānta* 3.10. The compound is geometrically explicit: the arm is the radius, the chord is the arm-string. The compound stem *dor-* is the visarga-sandhi form of the s-stem arm-noun *doṣ-*, exactly the form developed in *The Hand That Gives* §4.4. The *dor-jyā* attestation places *dóws- not only in linear-distance metrology but in geometric metrology: the arm is the radius of the trigonometric circle. The body-part heteroclite class generates measurement vocabulary not only for linear distance but for trigonometric ratio.

7.3 The Cross-IE Cubit Family

The cubit (and its kin: the ell, the half-cubit, the hand-breadth) is a near-universal IE metrological pattern, the body-part lexicalized as measurement unit across multiple branches:

- **Latin** *cubitus* “elbow; cubit” — the natural measure from elbow to fingertip
- **Greek** πῆχυς “forearm; cubit” — the same body-part-as-measure construction
- **Hebrew** *ammah* literally “mother of the arm” — the cubit-class measure explicitly anchored on the arm
- **Sumerian** *kuš* “cubit” — the same metrological category in the ancient Near East
- **Old English** *ell* “cubit, ell”; Old High German *elina* (whence English *elbow* < OE *eln-boga* “ell-bow”) — the metrological term feeding back into the body-part lexicon

The Sanskrit *dos* = 18 inches sits inside this pattern. The smaller palm/span-class measurements — the σπιθαμή of δάρις, the παλαιστή of δῶρον 2, Old Irish *derna*, Continental Celtic DERNV[— sit at the other end of the same body-part metrology. Together, hand and arm cover the relevant linear-distance scales: cubit (~18 inches), span (~9 inches), palm (~3 inches), digit (~3/4 inch). The body-part heteroclite class generates measurement vocabulary across the full range.

8. The Body-Part Heteroclite Class as the Productive PIE Pattern

The convergence of §§4–7 establishes that the body-part heteroclite class is a productive PIE pattern generating a coherent measurement-and-tool family in both members. The r-stem *de-r- produces palm/span-class linear measures across multiple branches (δῶρον 2, δάρις, *derna*, DERNV[, *dorē*), cupped-hand-volume measures

on zero-grade (δράξ, δράσσομαι, δράγμα, δραχμή), and scoop-and-ladle tool extensions in the eastern zone (*darvī*, *taruwal-*, *tarwāli-*, *targal/dgal*). The s-stem *dóws- produces cubit-class linear measures preserved most fully in Sanskrit (*dos*, *dor-jyā*) and paralleled across the cross-IE cubit family.

The two heteroclites are sister formations; the metrology and tool-extensions they generate are productive features of the class as a whole, not quirks of either member individually. This is typologically natural: body-part terms acquire measurement readings when the body part is habitually used as a measuring instrument, and tool-extensions when the body part is habitually used as a tool-prototype (Hilpert 2007; Pattillo 2014). The framework’s contribution is the recognition that the cross-IE T-R/D-R cluster — palm, handful, ladle, cobra-hood, cubit, sine — belongs to this productive pattern rather than being a scatter of isolated lexemes of unknown origin.

The body-schema of §3 provides the deeper unification. The cupped hand and the arm-and-cupped-hand configuration generate all three measurement registers (palm-linear, handful-volume, arm-cubit) and the tool-extension family (ladle as frozen arm-and-cupped-hand). Different daughter languages foreground different aspects of the same configuration, but the unified body-schema is recoverable through the cross-IE pattern.

9. The Etymology of Greek δῶρον 2

9.1 The Derivation

The framework derives δῶρον 2 from the consonant-stem r-stem *de-r- via Szemerényi’s Law (§2.1) followed by thematicization (§2.2):

Step	Form	Mechanism	Status
1	*do-r-s	o-grade r-stem with nom. sg. *-s	Standard PIE inflection
2	*dōr	Szemerényi’s Law: pre-PIE *-Vrs# → PIE *-Ũr#	Lundquist & Yates 2018: 2083; Szemerényi 1996: 113–119
3	*dōr-o-n	Thematicization of athematic r-stem; *-os/*-on extension	Standard PIE morphological process (Beekes 2011: 211–214; Fortson 2010: 132–134)
4	δῶρον	Greek surface phonology	Regular Greek; attested from Mycenaean <i>do-ra</i> (KN Sd 4401)

Table 1. The derivation of Greek δῶρον 2 from PIE *de-r-. Each step uses only sound rules and morphological processes uncontroversial in the standard handbook treatments of PIE.

Each step of the derivation uses only mechanisms uncontroversial in standard handbook treatments of PIE. Szemerényi’s Law on *-Vrs# appears in every IE textbook; its operation on *do-r-s requires no additional hypothesis. Thematicization of athematic r-stems with the *-os/*-om/*-on formation is documented in the standard reference grammars; its application to the Szemerényi-derived nominative *dōr is routine compositional morphology.

9.2 The Greek-Internal Model

The same morphological pattern that produces εὔπατορ (from *eu-patōr-s) and Homeric δώτορ “giver” (from *deh₃-tōr-s) also produces, one step further, δῶρον 2. In all three cases, Szemerényi’s Law on a consonant-

stem r-stem nominative produces the lengthened nominative. δῶτωρ and εὐπᾶτωρ surface as athematic nouns in -ωρ; the framework’s *dōr undergoes the additional step of thematization to surface as the neuter o-stem δῶρον. The thematization step is what distinguishes δῶρον 2 from its Greek-internal parallels; the Szemerényi mechanism is the same in all cases.

9.3 The Sibling Parallel with Albanian *dorë*

The framework’s *do-r-s → *dōr → *dōr-o-n for Greek δῶρον 2 and the companion paper’s *de-r-s → *dēr → *dēr-eh₂- → *dēr-ā → *dorë* for Albanian are structurally parallel derivations. Both begin from a consonant-stem r-stem nominative; both apply Szemerényi’s Law; both then extend the lengthened nominative with a derivational morpheme — thematic *-o-n for Greek, feminine *-eh₂- for Albanian. The differences are the ablaut grade (o-grade for δῶρον 2, e-grade for *dorë*) and the choice of extension. The convergence is morphological, not phonological: they share the same derivational scaffold applied in different grades.

9.4 The Sibling Parallel with δράξ on Zero-Grade

The δράξ family on zero-grade *dṛ-k- (§5) provides a Greek-internal sibling parallel that strengthens the morphological case considerably. The δῶρον 2 / δάρυς / δράξ family stands in Greek as a three-way ablaut alternation on the same r-stem base:

- o-grade *do-r-s → *dōr → δῶρον (thematicized)
- e-grade *de-r-i- → δάρυς (i-stem extension)
- zero-grade *dṛ-k- → δράξ (velar extension)

All three are productive Greek lexicalizations of the same root family, distinguished by ablaut grade and suffix selection. The claim that δῶρον 2 sits on *de-r- is confirmed language-internally by the zero-grade companion δράξ, which is morphologically unmistakable. No mainstream etymology accounts for the δωρ-/δαρ-/δραξ-pattern as a coherent family; the framework’s body-part-heteroclite reading is the only proposal that unifies them.

9.5 Mycenaean Evidence

The Linear B form *do-ra* (KN Sd 4401, in the chariot tablets) attests δῶρα as a thematicized neuter plural already in the thirteenth century BCE. The chariot-inventory context does not disambiguate between gift-δῶρα and palm-δῶρα, but it confirms that the thematicized neuter morphology was in place by the Mycenaean date — consistent with the framework’s PIE-date thematization.

9.6 The Greek Lexical Record for δῶρον 2

The body-part and metrological senses of δῶρον 2 are attested from the earliest Greek poetry forward:

- ***Iliad* Δ 109**: ἑκκαίδεκάδωρος “sixteen palm-widths long” — of Pandaros’s bow horns
- ***Hesiod Works and Days* 426**: δεκάδωρος “ten palm-widths” — of an axle
- ***Hesychian gloss*** δῶρον · παλαιστή — the body-part anchor in the lexicographic tradition
- ***Pollux Onomasticon* 2.144**: δῶρον as the front of the palm in a strictly anatomical context
- ***Miletus inscriptions***: δῶρον as a standard architectural unit

LSJ groups all of these under δῶρον II (palm-related), distinct from δῶρον I (gift). The framework’s etymology applies to all of these uses together, since they are all uses of the same lexeme.

9.7 What the Framework Supplies That Frisk and Beekes Left Open

Frisk marks δῶρον 2 “Origin: XX [etym. unknown]” while citing La Piana for the Albanian comparison. Beekes (2010: 363) marks it <?> and surveys rejected comparanda without a positive reconstruction. The framework supplies the full derivation both left open: Szemerényi’s Law on *do-r-s produces *dōr; thematization yields *dōr-o-n; the surface form is δῶρον 2, from *de-r- (the body-part heteroclite), not *deh₃- (the

give-verb). Frisk’s “Origin: XX” and Beekes’s “◁?▷” both reflect an etymology available through Szemerényi and thematicization, once the right root is identified.

10. δῶρον 1 “Gift” and the Vitruvian Folk Etymology

The framework adopts the standard derivation of δῶρον 1 “gift” from PIE *deh₃-ro- without modification. The reconstruction is uncontested, supported by an unbroken cross-branch cognate set: Greek δῶρον, Latin *dōnum* (from *deh₃-no-), Sanskrit *dāna-*, Armenian *tur*, OCS *darъ*, Old Irish *dán*, Welsh *dawn*. For δῶρον 1 specifically, the standard analysis stands.

Vitruvius (*De architectura* 2.3.3) tells us that the palm is called “gift” because giving is what the open hand does. The story is internally coherent — the HAND → GIVE grammaticalization pathway documented by Heine and Kuteva (2002) is real — but it is not derivational evidence. It is evidence that the homophony existed and was noticed by ancient writers, who did what speakers always do with homophones: constructed a semantic bridge to explain the apparent connection. English *ear* “organ of hearing” (from PIE *h₂ous-) and *ear* “spike of grain” (from PIE *h₂ek- “sharp”) are synchronically identical but etymologically unrelated; folk etymology links them. The same applies to the two δῶρα. The etymological histories are established by the comparative method; the folk etymology is a synchronic phenomenon that has no bearing on the historical reconstruction.

11. The Coexistence Reading

The framework does not claim exclusive rights to the IE hand-vocabulary. PIE *ǵ^hesr-/or-/to- (Pokorny IEW 447) is accepted in full as the source of the cross-IE “hand-as-grasp” word: Greek χεῖρ, Hittite *kessar*, Tocharian A *tsar* / B *ṣar*, Armenian *jeṛn*; via *ǵ^hesto-, Sanskrit *hásta-*, Avestan *zasta-*, Old Persian *dasta-*. PIE *deh₃- is accepted in full as the source of the give-verb and the gift-noun cluster on *deh₃-ro-/no-. PIE *d^her-/d^hergh- “hold firm, bind tight, knot” is accepted as a separate root generating the fist-cluster (Celtic *durno-, OIr *dorn*, Welsh *dwrn*, Breton *dorn*, Latvian *dūre*) and the broader hold-cluster developed in *A Firm Hand*.

The framework claims only that *de-r- is a distinct PIE etymon, centered on the cupped hand and the arm-and-cupped-hand body-configuration, with its own coherent derivational outputs in three measurement registers and a scoop-tool family. The four etyma likely had specialized senses in early PIE: *ǵ^hesr- the hand as joint/grasp; *de-r- the cupped hand and palm as body-configuration; *deh₃- the verbal act of extension-toward-recipient; *d^her-/d^hergh- the firm-holding-and-binding action of the closed hand.

Pokorny kept *der- (IEW 203), *dóws- (IEW 226), *ǵ^hesor- (IEW 447), *d^her- (IEW 252), and the give-verb (IEW 223–226) as separate entries; this paper is methodologically aligned with that taxonomy. Readers interested in the fist-cluster should consult *A Firm Hand*; the present paper concerns the cupped hand and the palm-and-measure family on *de-r-.

12. The Late-Twentieth-Century Departures

Three departures targeted the Pokorny IEW 203 grouping. Each is addressed in turn.

12.1 Schrijver 1995 on the δῶρον-/δαρον- “Obscurity”

Schrijver (1995: 58) declared the δῶρον-/δαρον- relation “obscure.” He was right that no straightforward sound rule connects δῶρον- and δαρον-. The framework does not dispute this. What it disputes is the assumption that the two forms stand in direct phonological relationship. They are sibling derivations on a shared r-stem base, not phonological variants of a single word — a morphological relationship that Schrijver’s question was not designed to capture.

With the addition of zero-grade δρᾱξ in §5, the three-way ablaut pattern becomes explicit: δωρ- / δαρ- / δρᾱξ- as the three ablaut-grade outputs of *de-r- with different suffixes. The “obscure” verdict was the right answer to the wrong question.

12.2 Beekes 2010 / Matasović 2009 on Celtic *u

These objections target the Celtic fist-cluster (*durno-, *dorn/dwrn*, Breton *dorn*, Latvian *dūre*), not the open-palm or palm-and-measure cluster. The framework reassigns the fist-cluster to PIE *d^her-/ *d^hergh- in *A Firm Hand*, with the Celtic *u-vocalism falling out as the u-grade of *d^her- (parallel to Sanskrit *dhūr-* “yoke” < *d^hur-h₁-). A fist is the hand in its firm-knotted posture; the etymological homeland of the Celtic fist-word is the firm-holding cluster, not the open-palm cluster on *de-r-.

This reassignment removes the Celtic *u problem from the present paper entirely. What remains for *de-r- is the palm-and-measure cluster proper — δῶρον 2, δᾱρίς, *derna*, DERNV[, *dorë*, the zero-grade δρᾱξ family, and the eastern-zone scoop-tool family — all on the e-grade, o-grade, or zero-grade of *de-r- without any *u-vocalism.

12.3 The “Substrate” Pattern

Multiple components of the cross-IE T-R/D-R cluster have been classified as substrate borrowings in mainstream IE etymology: Pre-Greek for δρᾱξ; non-IE loanword for the Anatolian *taruwal-/tarwāli-* family; uncertain origin for Armenian *targal/dgal*. The recurrence of this attribution across forms that share a tight semantic field and a consistent consonant skeleton is itself structural evidence that mainstream PIE phonology has not adequately reduced the cluster to a single root. The framework’s reading — that these forms are productive outputs of *de-r- with the cupped-hand body-schema as the unifying semantic anchor — provides a positive IE etymology for forms that mainstream treatments have had to file under “substrate” precisely because the cluster’s coherence resisted reduction to any single standard PIE entry.

13. Scope and Falsifiability

Scope

The framework does not introduce a new PIE root. The body-part heteroclite *de-r- is developed in *The Hand That Gives*; the present paper develops the measurement-and-tool cluster issuing from it. The standard derivation of δῶρον 1 from *deh₃-ro- is preserved without modification. The Celtic fist-cluster is reassigned to *d^her-/ *d^hergh- in *A Firm Hand* and is outside the scope of this paper. Germanic *trog* and Old Irish *drochta* “wooden basin” are left to their mainstream treatments on PIE *deru-/ *drū-.

Falsifiability

The framework is falsifiable in five distinct ways.

1. A positive demonstration that δῶρον 2 “palm” descends from *deh₃- by some derivational path (heteroclitic apparatus, semantic extension from gift to palm, or alternative morphology) would remove the framework’s central contribution at §9.
2. Demonstrated **within-branch variation in the δωρ-/δαρ-/δρᾱξ- distribution** not captured by the sibling-derivation analysis (§9.4) would reopen Schrijver’s “obscure” verdict.
3. A **positive Anatolian etymology for taruwal-/tarwāli-** derived from a non-*de-r-/non-*deh₃- root with independent Anatolian-internal support would weaken the cross-branch coherence of the cluster, though it would not directly affect the §9 derivation of δῶρον 2.
4. Demonstration that the **Sanskrit dos = 18 inches and dor-jyā = sine** attestations belong to a different etymological root from the s-stem arm-noun *doṣ-/doṣān-* would weaken the §7 arm-as-measure parallel, though it would not directly affect the §9 derivation.

5. Demonstration that the **δράξ family descends cleanly from a Pre-Greek substrate** source with verifiable substrate-language attestations would weaken the framework’s §5 zero-grade extension, though the morphological pattern would remain available as a parallel structural argument.

14. Probability Assessment

The central thesis of this paper — that Greek δῶρον 2 descends from *de-r- via Szemerényi plus thematization; that it belongs with δάρις, *derna*, DERNV[, Albanian *dorë*, and zero-grade δράξ in a body-part-and-measure cluster on *de-r-; that the s-stem sister *dóws- generates a parallel arm-as-cubit cluster; and that the eastern-zone ladle/scoop cluster belongs to the same root via the PALM/CUPPED-HAND → SCOOP/LADLE extension — is assessed at approximately **50–60%**, conditional on Paper 1’s framework.

The conditional probability sits in the middle-to-upper range of the disputed-but-plausible interval because the framework integrates relatively secure material (the Szemerényi+thematicization derivation of δῶρον 2, supported by the εὐπάτωρ/δῶτωρ Greek-internal pattern) with more contested material (the eastern-zone ladle-cluster reassignment from *dōru “wood” to *de-r-, and the body-schema unification of three measurement registers under one root). Readers who accept the δῶρον 2 etymology while reserving judgment on the ladle-cluster can still extract a coherent framework for the palm-as-measure cluster.

In Favor

- **Frisk and Beekes already treat δῶρον 1 and δῶρον 2 as separate entries.** The two-entry treatment is not novel; what is novel is the etymology supplied for the second.
- **The morphological apparatus** — Szemerényi’s Law on consonant-stem r-stem nominatives, followed by thematization — uses only standard handbook tools (Lundquist and Yates 2018: 2083; Szemerényi 1996: 113–119; Beekes 2011: 211–214; Fortson 2010: 132–134).
- **The δωρ-/δαρ-/δραξ- ablaut family** is morphologically coherent across all three ablaut grades of *de-r-, with zero-grade δράξ supplying language-internal Greek confirmation of the framework’s root identity.
- **The δράξ family** provides a Greek-internal volumetric-measure register parallel to δῶρον 2’s linear-measure register — two specializations of the same body-schema that fit cross-linguistic body-part metrology naturally.
- **The arm-as-measure parallel on *dóws-** (Sanskrit *dos* = 18 inches, *dor-jyā* = sine) places the cluster inside a productive PIE body-part-as-measure pattern.
- **The cross-IE cubit family** (Latin *cubitum*, Greek πῆχυς, Hebrew *ammah*, Old English *ell*) is a near-universal IE metrological pattern; the Sanskrit *dos* sits inside it.
- **The cobra-hood diagnostic** in Sanskrit *darvī* — applying the same word to both ladles and the splayed hood of a cobra — is the framework’s strongest single piece of internal-IE evidence. The standard *dāru- “wood” etymology cannot reach the cobra-hood sense; the configuration-channel reading on *de-r- handles both senses through a single shape-based mechanism.
- **The parallel Sanskrit hand-as-spoon compounds** (*dāruhasta-* “wood-hand,” *dāruhastaka-*, *pāṇikā* from *pāṇi-* “hand”) provide typological support for the body-schema reading. Sanskrit lexicalizes spoons compositionally from body-part terms running alongside *darvī*’s configuration-channel lexicalization, documenting that the cupped-hand-to-scoop conceptual link was lexically active in Sanskrit.
- **The Indic-Anatolian-Armenian tool-cluster** is geographically and phonologically coherent.
- **Mycenaean do-ra** (KN Sd 4401) confirms the thematicized neuter morphology was in place by the thirteenth century BCE.

Against

- **The two-δῶπα analysis remains a minority view.** The post-Schrijver consensus has favored treating the gift-side and palm-side as derivationally unified, or the palm-side as a secondary development from the gift-side.
- **The body-part heteroclite class** as a productive PIE pattern generating three measurement registers and a scoop-tool family is not standard in current handbook treatments; Pokorny kept *der- and *dóws- as separate entries without explicitly classifying them as sister heteroclites with productive cross-register output.
- **The eastern-zone ladle-cluster reassignment** from mainstream *dóru “wood” to framework *de-r- is the framework’s most contested claim. The cobra-hood and *dāruhasta*- arguments are strong, but specialists committed to the wood-etymology will resist.
- **The δρᾶξ family’s reassignment** from Beekes’s Pre-Greek substrate to zero-grade *d̥r̥k- of *de-r- is similarly contested. The α-vocalism is consistent with either reading; the framework’s analysis has the advantage of explaining the semantic clustering, but contests an established specialist position.
- **The chief residue** is the irregular plural *duar* of Albanian *dorë* (treated in the standalone *dorë* paper), which no current reconstruction handles transparently.

Project-Level Marginal Probability

The marginal probability of the broader project — integrating conditional probabilities across *The Hand That Gives*, the standalone Albanian *dorë* paper, *A Firm Hand*, and the present paper — sits in the **2–5%** range. Much of the evidence supports multiple strata simultaneously rather than each stratum independently; the honest marginal range is toward the lower end. We regard the central claim as more likely wrong than right at the marginal level, and the evidential structure as sufficiently dense and sufficiently testable to warrant publication for scholarly examination.

15. Authorship and Contributions

This paper is co-authored by **J.M. Chavis (human)** and **Claude (Anthropic AI; primarily Claude Opus 4.7)**.

Chavis’s contributions. The central hypothesis originates with Chavis: the identification of the dual attestation of Greek δῶπον — gift and palm-breadth — as evidence of two etymologically distinct words was the project’s founding insight. Chavis directed the research agenda at every stage: the architectural decision to split the previous combined Paper 1 into two papers (*The Hand That Gives* on the arm-extension and Anatolian material; the present companion on δῶπον 2 and the body-part-and-measure cluster); the inclusion of the s-stem *dóws- arm-as-measure material as parallel to the r-stem palm-as-measure cluster; the redirection of the Albanian *dorë* defense to its own standalone paper; the reframing of the Celtic fist-cluster onto the *d^her-/*d^hergh- foundation in *A Firm Hand*; the methodological pivot to the coexistence reading; the recognition that the cupped hand (rather than the splayed open hand) is the framework’s primal body-schema anchor; the identification of δρᾶξ as the zero-grade volumetric-measure register; the integration of the eastern-zone ladle-cluster (*darvī*, *taruwal*-, *tarwāli*-, *targal/dgal*) on *de-r- via the PALM/CUPPED-HAND → SCOOP/LADLE extension; and the identification of the logical flaw in the original §6.2 *dāruhasta*- tautology argument, which led to the body-part-channel / configuration-channel distinction articulated in §3.1 and applied throughout §6. All fundamental decisions about scope, framing, and direction rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude’s contributions are primarily in source synthesis, exposition, and analytical development. Claude carried out the literature reading, cross-branch corroboration searches, and technical

drafting under Chavis’s direction. Claude engaged with Lundquist and Yates 2018, Szemerényi 1996, Beekes 2011, Fortson 2010, Meier-Brügger 2003, Hilpert 2007, Pattillo 2014, Stifter 2022, Pedersen 1913, Chantraine 1968, Frisk 1960–72, Mayrhofer EWAia, Martirosyan 2010, and the Sanskrit lexicographical and metrological sources (*Mātaṅgalīlā*, *Sūryasiddhānta*, Cologne lexicons, wisdomlib); examined Pokorny IEW pp. 203, 226, 252, and 447 directly; verified the Mycenaean Linear B attestation *do-ra* (KN Sd 4401); verified Xenophon *Anabasis* 1.5.6 for the Persian *kapithi* measure; and integrated the architectural alignment with the companion papers. The probability estimates are Claude’s own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management.

Both authors take responsibility for the framework and the specific claims made in this paper.

Select References

- Beekes, R.S.P. 2010. *Etymological Dictionary of Greek*. 2 vols. Leiden: Brill.
- Beekes, R.S.P. 2011. *Comparative Indo-European Linguistics: An Introduction*. 2nd ed. Amsterdam: Benjamins.
- Chantraine, P. 1968. *Dictionnaire étymologique de la langue grecque*. Paris: Klincksieck.
- Chavis, J.M. and Claude. 2026a. *The Hand That Gives: Pre-PIE *de- as Arm-Extension, the Directional Particle *de/do, the Body-Part Heteroclite Arm-Noun, and the Give-Verb *deh₃-*. Paper 1 of the der- project.
- Chavis, J.M. and Claude. 2026c. *The Etymology of Albanian dorë*. Standalone companion paper.
- Chavis, J.M. and Claude. 2026d. *A Firm Hand: PIE *d^her-/ *d^hergh- “Hold Firm, Bind Tight, Knot” and the Celtic *durno- Fist-Cluster*. Companion paper (in preparation).
- Demiraj, B. 1990. “Sull’etimologia di alb. dorë.” *Indogermanische Forschungen* 95: 99–110.
- Demiraj, B. 1997. *Albanische Etymologien*. Amsterdam: Rodopi.
- Fortson, B.W. 2010. *Indo-European Language and Culture*. 2nd ed. Oxford: Wiley-Blackwell.
- Frisk, H. 1960–72. *Griechisches Etymologisches Wörterbuch*. Heidelberg: Winter.
- Heine, B. and T. Kuteva. 2002. *World Lexicon of Grammaticalization*. Cambridge: Cambridge University Press.
- Hilpert, M. 2007. “Chained metonymies in lexicon and grammar.” In G. Radden et al. (eds.), *Aspects of Meaning Construction*, 77–98. Amsterdam: John Benjamins.
- Lundquist, J. and A.D. Yates. 2018. “The Morphology of Proto-Indo-European.” In Klein, J. et al. (eds.), *Handbook of Comparative and Historical Indo-European Linguistics*, vol. 3, 2079–2195. Berlin: De Gruyter.
- Marstrand, C. 1919. *Caractère indo-européen de la langue hittite*. Christiania: Dybwad.
- Martirosyan, H. 2010. *Etymological Dictionary of the Armenian Inherited Lexicon*. Leiden: Brill.
- Matasović, R. 2009. *Etymological Dictionary of Proto-Celtic*. Leiden: Brill.
- Mayrhofer, M. 1992–96. *Etymologisches Wörterbuch des Altindoarischen (EWAia)*. Heidelberg: Winter.
- Meier-Brügger, M. 2003. *Indo-European Linguistics*. Berlin: De Gruyter.
- Pattillo, K. 2014. *Cross-Linguistic Metonymies in Human Limb Nomenclature*. PhD dissertation, University of Wisconsin-Milwaukee.

- Pedersen, H. 1909, 1913. *Vergleichende Grammatik der keltischen Sprachen*. Göttingen: Vandenhoeck & Ruprecht.
- Pokorny, J. 1959. *Indogermanisches etymologisches Wörterbuch (IEW)*. Bern: Francke.
- Schrijver, P. 1995. *Studies in British Celtic Historical Phonology*. Amsterdam: Rodopi.
- Stifter, D. 2022. “Continental Celtic DERNV[at the Bronze of Novallas.” *Études Celtiques* 48: 41–60.
- Szemerényi, O.J.L. 1996. *Introduction to Indo-European Linguistics*. 4th ed. Oxford: Oxford University Press.
- Vitruvius. *De architectura*. (1st century BCE.)
- Xenophon. *Anabasis*. (4th century BCE.)